COURSE SYLLABUS

1. COURSE TITLE

Numerical and Simulation Techniques in Finance
金融中的数值与模拟方法

2. COURSE CODE

FINM4053

3. PRE-REQUISITE

(STAT2063 Probability Theory or STAT2023 Advanced Probability), and
MATH2023 Ordinary and Partial Differential Equations, and
(FINM4073 Advanced Financial Mathematics or FINM4063 Stochastic Calculus for Finance or FINM4003 Financial Mathematics II (for FM Students)), and
COMP3153 C++ Programming Language

4. CO-REQUISITE

Nil

5. NO. OF UNITS

3

6. CONTACT HOURS

42

7. MEDIUM OF INSTRUCTION (MOI)

English

8. OFFERING UNIT

Financial Mathematics Programme, Faculty of Science and Technology.

9. SYLLABUS PREPARED & REVIEWED BY

Prepared by: Dr. Nicolas Langrené
Reviewed by: Dr. Peng Jin

10. AIMS & OBJECTIVES

1. To introduce students to the basic knowledge about solving and implementing

numerical problems of increasing complexity that arise in finance.

2. To familiarize students with code design that reflects the structure of numerical problems from finance.

3. To apply Monte Carlo simulation in option pricing.

11. COURSE CONTENT

1. Numerical computation
   a) Computer arithmetic
   b) Measuring approximation error
   c) Numerical instability and ill conditioning
   d) Complexity of algorithms
   e) Python modules for numerical computation
2. Finite difference method
   a) Classification of differential equations
   b) Numerical differentiation
   c) Finite difference solution
   d) Numerical stability of PDE schemes
   e) Coordinate transformation
3. Univariate simulation
   a) Congruential generators
   b) Inverse transform sampling
   c) Heavy-tailed distributions
4. Multivariate simulation
   a) Multivariate distributions
   b) Simulate correlated random variables
   c) Copulas
5. Monte Carlo methods
   a) Brownian motion simulation
   b) Simulation of stochastic differential equations
   c) Monte Carlo option pricing
6. Advanced Monte Carlo (time permitting)
   a) Variance reduction
   b) Rare event simulation
   c) Option Greeks simulation

12. COURSE INTENDED LEARNING OUTCOMES (CILOS) WITH MATCHING TO PILOS

Programme Intended Learning Outcomes (PILOs)

Programme Title: Bachelor of Science (Honours) in Financial Mathematics	
PILO	Upon successful completion of the Programme, students should be able to:
PILO 1	Apply the basic principles of financial mathematics to clearly
PILO 2	explain the phenomena and problems in financial markets;
PILO 3	Interpret quantitative models for pricing derivatives, managing trading
PILO 4	Use mathematical and financial software to perform computation of financial data analysis and pricing and valuation of financial instruments;
PILO 5	Conduct independent research in financial management and exploring business opportunities in financial markets;
PILO 6	Develop the capacity to work as part of a team.

CILOs-PILOs Mapping Matrix

Course Code & Title: FINM4053 Numerical and Simulation Techniques in Finance		
CILO	**Upon successful completion of the course, students should be able to:**	**PILO(s) to be addressed**
CILO 1	Explain basic concepts of numerical analysis and Monte Carlo simulation	PILO 2
CILO 2	Analyze problems and design numerical and simulation algorithms	PILO 3
CILO 3	Apply numerical and simulation techniques in financial and economic models	PILO 4

13. TEACHING & LEARNING ACTIVITIES (TLAS)

CILO No.	**TLAs**
CILO 1	• **Lectures:** Three hours of lecture will be given per week. This is a standard course on numerical methods with financial applications. Instructor will explain the concept, definition, and theory of numerical computation to students • **Tutorials:** One hour of tutorials will be given per week. Assignments will be given to students to strengthen their concepts. Solutions to assignments will be given and discussed

	in the tutorial class. • **Assignments and In-class exercises:** Assignments and in-class exercises are designed to measure the students' understanding of numerical computation, numerical solution of differential equations, and Monte Carlo simulation methods.
CILO 2	• **Lectures**: Three hours of lectures and one hour of tutorials will be given per week. This is a standard course on numerical methods with financial applications. Instructor will show examples to students to illustrate the evaluation techniques. • **Tutorials:** The tutor will explain the evaluation in more details. • **Assignments and In-class exercises:** Assignments and in-class exercises are designed to measure the students' understanding of numerical computation, numerical solution of differential equations, and Monte Carlo simulation methods.
CILO 3	• **Lectures:** Three hours of lectures and one hour of tutorials will be given per week. This is a standard course on numerical methods with financial applications. Instructors will show examples to students to illustrate some real financial pricing and optimization problems, and instructors will guide students to solve problems. • **Tutorials:** The tutor will explain the solution in more details. • **Assignments and In-class exercises:** Assignments and in-class exercises are designed to measure the students' understanding of numerical computation, numerical solution of differential equations, and Monte Carlo simulation methods.

14. ASSESSMENT METHODS (AMS)

Type of Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
Assignments	15%	CILO 1, 2	Assignments are designed to measure students' skills in solving problems.
Project	20%	CILO 1, 3	Projects are designed to apply numerical and simulation methods on financial models.
In-class exercises	25%	CILO 1, 2	In-class exercises are conducted to measure students' understanding of some course materials in a timely manner.

Final Exam	40%	CILO 1-3	The final examination is designed to see how far students have achieved the intended learning outcomes. Students will need to have a thorough understanding of the knowledge and skills, and apply them correctly in different contexts well.

15. TEXTBOOKS / RECOMMENDED READINGS

Textbooks:

1. M. Gilli, D. Maringer, and E. Schumann. *Numerical methods and optimization in finance*. 2nd ed. Academic Press, 2019.

Recommended Readings:

1. R. Korn, E. Korn, and G. Kroisandt. *Monte Carlo methods and models in finance and insurance*. CRC press, 2010.
2. D. Duffy. *Finite difference methods in financial engineering: a partial differential equation approach*. John Wiley & Sons, 2013.
3. Y. Hilpisch. *Python for finance: mastering data-driven finance*. O'Reilly Media, Inc. 2nd ed. 2018.
4. S. Shreve. *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer, 2004.
5. N. J. Higham. *Accuracy and stability of numerical algorithms*. 2nd ed. 2002.
6. J. Bouchaud and M. Potters. *Theory of financial risk and derivative pricing: from statistical physics to risk management*. 2nd. Cambridge University Press, 2003.
7. P. Kloeden, E. Platen, and H. Schurz. *Numerical solution of SDE through computer experiments*. Springer Science & Business Media, 2012.
8. Paul Glasserman, Monte Carlo Methods in Financial Engineering, Springer, 2003.

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